AAE-E2001 - Computational Fluid Dynamics D

Assignment 2: Lid-driven cavity

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1 Theory Exercise

The 1D convection diffusion equation is:

$$\frac{\partial c}{\partial t} + u\frac{\partial c}{\partial x} = \Gamma\frac{\partial^2 c}{\partial x^2} \tag{1}$$

Now the conservative form of this equation would be:

$$\frac{\partial c}{\partial t} = -\frac{\partial f_C}{\partial x} + \frac{\partial f_D}{\partial x} \tag{2}$$

where the convective and diffusive fluxes are:

$$f_C = uc, \quad f_D = \Gamma\frac{\partial c}{\partial x} \tag{3}$$

with constant u, Γ and Δx. By considering a control volume (cell) i of width Δx and cross-sectional area S, the volume would be $V_i = \Delta x\ S$. By integrating Eq. 2 over the whole control volume, we have:

$$\int_V \frac{\partial c}{\partial t} dV = -\int_V \frac{\partial f_C}{\partial x} dV + \int_V \frac{\partial f_D}{\partial x} dV \tag{4}$$

Here, the Gauss theorem is applied to get surface integrals,

$$\int_V \frac{\partial c}{\partial t} dV = -\int_S f_C \vec{n}.dS + \int_S f_D \vec{n}.dS \tag{5}$$

Using the divergence theorem in 1D, the volume integrals of the flux gradients turn into differences of the fluxes at the cell faces $i \pm 1/2$.

$$\int_V \frac{\partial c}{\partial t} dV = -(f_{C_{i+1/2}} - f_{C_{i-1/2}})S + (f_{D_{i+1/2}} - f_{D_{i-1/2}})S \tag{6}$$

Approximating the cell average of c by its value at the cell center, we have:

$$\int_{V_i} \frac{\partial c}{\partial t} dV \approx V_i \frac{\partial c_i}{\partial t} = \Delta x\ S\ \frac{\partial c_i}{\partial t} \tag{7}$$

substituting Eq. 7 into Eq. 6 and dividing by $V_i = \Delta x\ S$ yields the semi discrete finite volume equation:

$$\frac{\partial c_i}{\partial t} = \frac{-(f_{C_{i+1/2}} - f_{C_{i-1/2}}) + (f_{D_{i+1/2}} - f_{D_{i-1/2}})}{\Delta x} \tag{8}$$

We now approximate the face fluxes using midpoint rule for the integral, linear interpolation for c at the faces, and second order central differences for the diffusion term. So we would have:

$$\begin{aligned} f_{C_{i+1/2}} &= (uc)_{i+1/2} = u\, c_{i+1/2} = u\frac{c_i + c_{i+1}}{2}, \\ f_{C_{i-1/2}} &= (uc)_{i-1/2} = u\, c_{i-1/2} = u\frac{c_{i-1} + c_i}{2} \end{aligned} \tag{9}$$

Now in order the approximate the diffusive fluxes, the gradients are approximated by central differences at cell faces:

$$(\frac{\partial c}{\partial x})_{i+1/2} \approx \frac{c_{i+1} - c_i}{\Delta x}, \qquad (\frac{\partial c}{\partial x})_{i-1/2} \approx \frac{c_i - c_{i-1}}{\Delta x} \tag{10}$$

Thus,

$$\begin{aligned} f_{D_{i+1/2}} &= \Gamma(\frac{\partial c}{\partial x})_{i+1/2} = \Gamma\frac{c_{i+1} - c_i}{\Delta x} \\ f_{D_{i-1/2}} &= \Gamma(\frac{\partial c}{\partial x})_{i-1/2} = \Gamma\frac{c_i - c_{i-1}}{\Delta x} \end{aligned} \tag{11}$$

Of course in these equations we assumed that u is positive. Now by substituting these values into Eq. 8 we would have:

$$\frac{\partial c_i}{\partial t} = \frac{-(u\frac{c_i+c_{i+1}}{2} - u\frac{c_{i-1}+c_i}{2}) + (\Gamma\frac{c_{i+1}-c_i}{\Delta x} - \Gamma\frac{c_i-c_{i-1}}{\Delta x})}{\Delta x} \tag{12}$$

Rearranging this,

$$\frac{\partial c_i}{\partial t} = -\, u\frac{c_{i+1} - c_{i-1}}{2\Delta x} + \Gamma\frac{c_{i+1} - 2\, c_i + c_{i-1}}{(\Delta x)^2} \tag{13}$$

which is exactly the same discretization obtained by the **finite difference** form.

2 Simulation exercise

In this exercise, a lid-driven cavity flow is simulated. The case consists of an incompressible fluid confined in a square cavity, where the motion of the top wall (the lid) drives the flow. To enable comparison of the results, the characteristic velocity U, cavity length L, and kinematic viscosity ν are chosen such that the Reynolds number is

$$\mathrm{Re} = \frac{UL}{\nu} = 100,$$

which corresponds to laminar flow conditions.

U	1 (m/s)
L	1 (m)
ν	0.01 (m^2/s)

Table 1: Chosen values for the lid-driven cavity simulation.

Table 1 summarizes the parameters used in the simulations. The solver computes the velocity and pressure fields in the cavity. A no-slip boundary condition is applied for the velocity on all solid walls, while the top lid moves with a prescribed velocity of $U = 1$ m/s. For the pressure, a zero-gradient boundary condition is imposed at the walls. The initial condition for the flow is a quiescent fluid, i.e. the velocity is set to zero everywhere inside the cavity at $t = 0$.

The simulation is performed with three different mesh resolutions, *i.e.* $n_x \times n_y = 10 \times 10$, 40×40, and 80×80. The Courant number,

$$\mathrm{C_o} = u\frac{\Delta t}{\Delta x},$$

depends on the local cell size and velocity. For the present laminar case, the velocity in the cavity does not exceed the lid velocity, so the values in Table 2 are chosen to keep the Courant number approximately constant at Co ≈ 0.1 for all meshes, enabling a consistent comparison.

$n_x \times n_y$	$\Delta x = \dfrac{L_x}{n_x}$	Δt
10×10	0.1	0.01
40×40	0.025	0.0025
80×80	0.0125	0.00125

Table 2: Time-step values corresponding to a Courant number of approximately Co ≈ 0.1.

Figure 1 shows the computational grids used in the simulations. The left panel corresponds to the coarsest mesh (10×10), while the right panel shows the finest mesh (80×80).

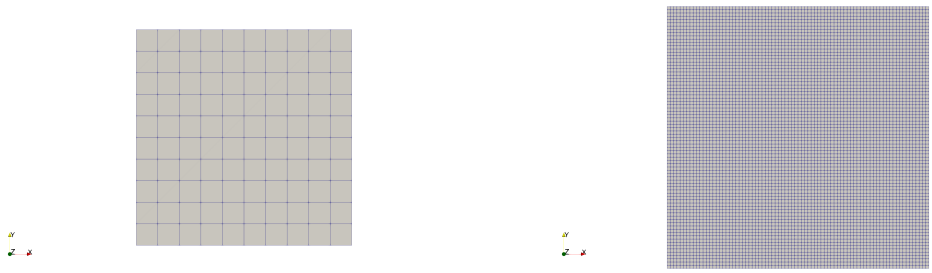

Figure 1: Computational meshes for the 10×10 (left) and 80×80 (right) grid resolutions.

The simulations were run for a total of ten by-pass time, $T = 10L/U = 10$ s. Figure 2 shows the flow field at $t = 10$ s for the three different mesh resolutions. The flow field is dominated by a large primary vortex in the upper part of the cavity, driven by the moving lid. In addition, small secondary vortices form in the two bottom corners where the flow is forced to turn sharply. These corner vortices are only weakly captured on the 10×10 grid but become progressively sharper and better resolved on the 40×40 and 80×80 meshes, reflecting the improved resolution of the near-wall shear layers.

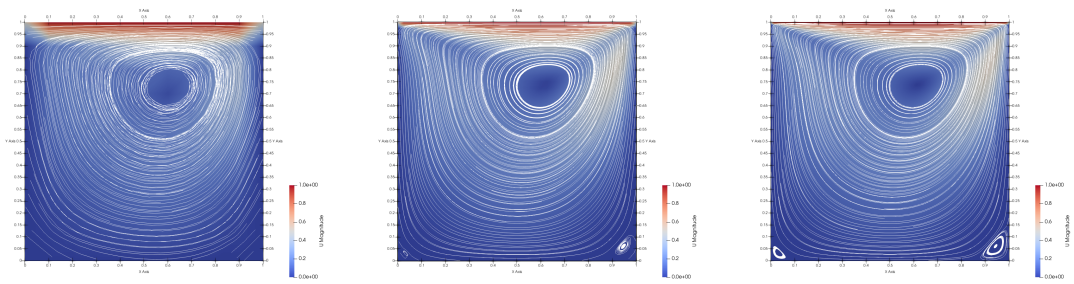

Figure 2: Streamline patterns at $t = 10$ s for the three grid resolutions: 10×10 (left), 40×40 (center), and 80×80 (right).

For the 40×40 grid, the horizontal velocity component u_x is evaluated along the vertical centerline of the cavity ($x = 0.5L$). Figure 3 shows u_x as a function of y at $t = 1, 2, 3, 4$, and 10 s, together with the benchmark steady-state data of Ghia *et al.* [1]. As time progresses, the transient profiles move towards the reference curve and gradually converge to the steady-state solution. The final profile at $t = 10$ s agrees well with the data of Ghia *et al.*.

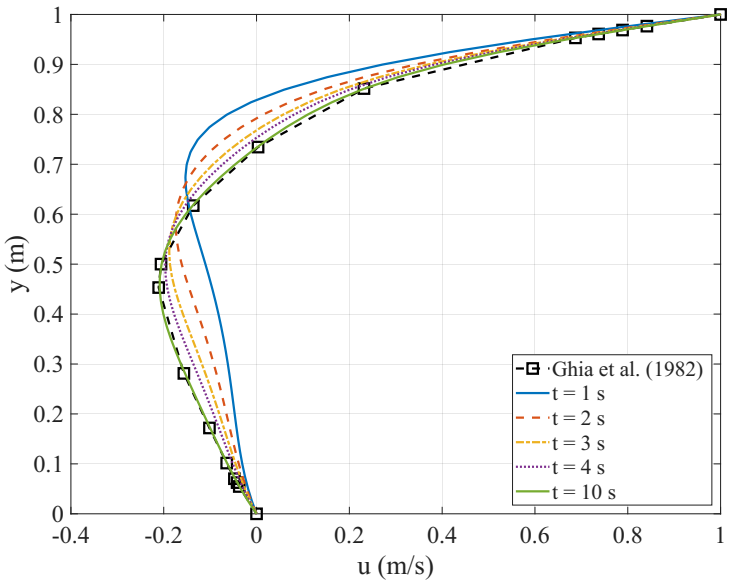

Figure 3: Horizontal velocity u_x along the vertical centerline ($x = 0.5L$) for the 40×40 grid at $t = 1, 2, 3, 4$, and 10 s, compared with the steady-state reference data of Ghia *et al.* [1].

For the 40×40 grid, the vertical velocity component v is evaluated along the horizontal centerline of the cavity ($y = 0.5L$). Figure 4 shows v as a function of x at $t = 1, 2, 3, 4$, and 10 s, together with the benchmark steady-state data of Ghia *et al.* [1]. As time advances, the transient profiles gradually approach the reference curve: the locations and magnitudes of the peak upward and downward velocities converge towards the Ghia *et al.* solution. By $t = 10$ s, the numerical profile is in close agreement with the benchmark data over most of the cavity, with only small discrepancies near the side walls, again attributable to the finite grid resolution.

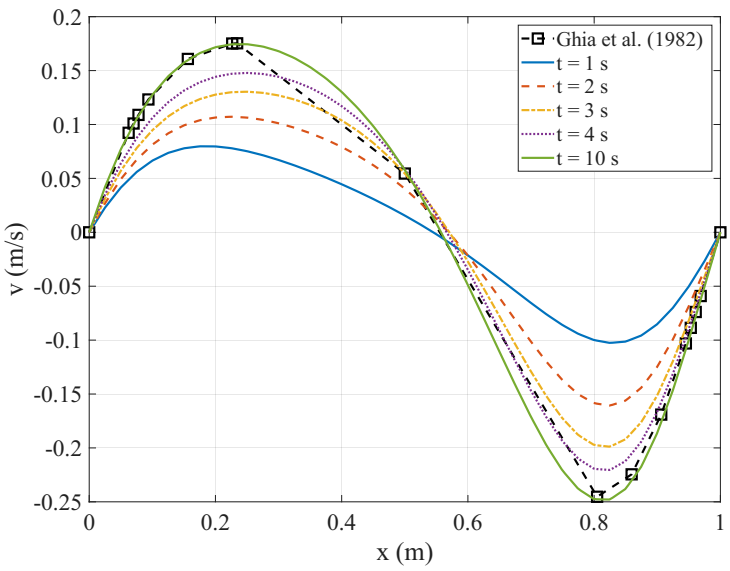

Figure 4: Vertical velocity v along the horizontal centerline ($y = 0.5L$) for the 40×40 grid at $t = 1, 2, 3, 4$, and 10 s, compared with the steady-state reference data of Ghia *et al.* [1].

At the final time $t = 10$ s, the horizontal velocity component u_x is evaluated along the vertical centerline of the cavity ($x = 0.5L$) for all three mesh resolutions. Figure 5 compares these profiles with the benchmark data of Ghia *et al.* [1]. The coarsest grid (10×10) shows noticeable deviations in the core region of the cavity, especially near the velocity minimum. As

the mesh is refined to 40×40 and 80×80, the numerical profiles collapse onto each other and match the reference curve very closely over the entire height, indicating grid convergence and showing that the medium and fine meshes are sufficient to capture the main flow features at this Reynolds number.

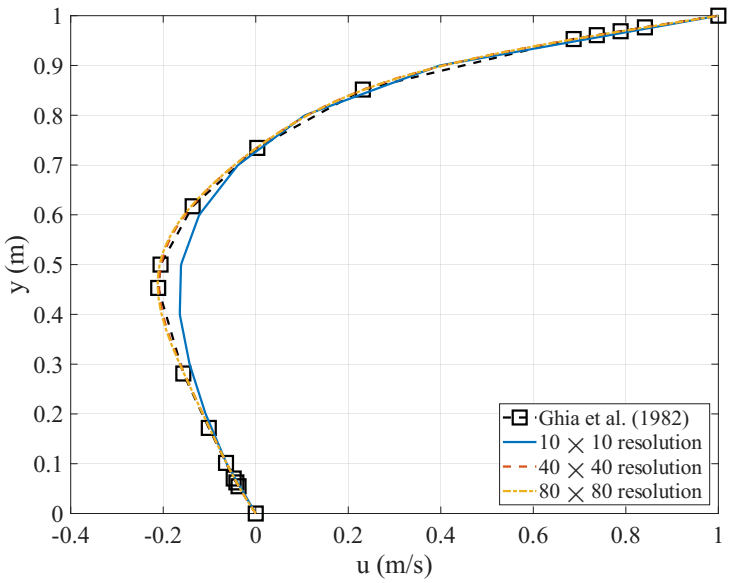

Figure 5: Horizontal velocity u_x along the vertical centerline ($x = 0.5L$) at $t = 10$ s for the three grid resolutions (10×10, 40×40, and 80×80), compared with the steady-state reference data of Ghia *et al.* [1].

At the final time $t = 10$ s, the vertical velocity component v is sampled along the horizontal centerline of the cavity ($y = 0.5L$) for all three mesh resolutions. Figure 6 compares these profiles with the benchmark data of Ghia *et al.* [1]. The 10×10 grid shows noticeable errors, in particular an underprediction of the positive peak near the left wall and a shifted minimum in the right half of the cavity. Refining the mesh to 40×40 and 80×80 yields profiles that almost coincide with each other and closely follow the reference curve, indicating that the vertical-velocity distribution is well resolved on the medium and fine meshes.

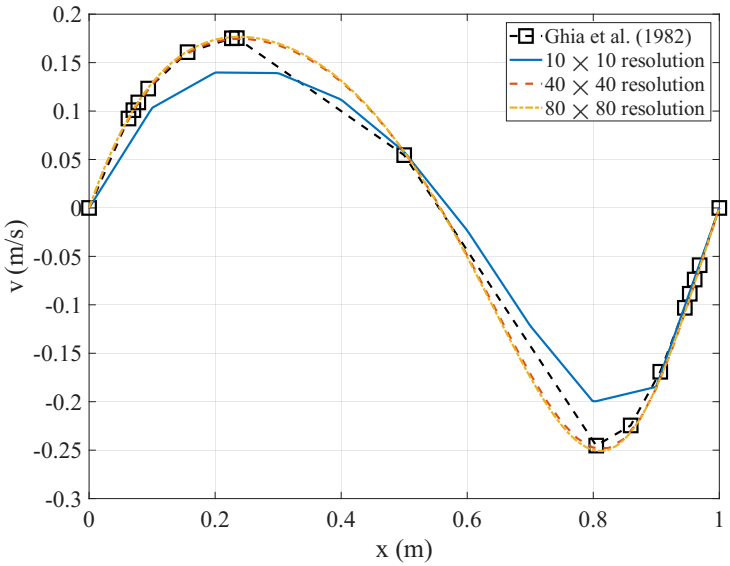

Figure 6: Vertical velocity v along the horizontal centerline ($y = 0.5L$) at $t = 10$ s for the three grid resolutions (10×10, 40×40, and 80×80), compared with the steady-state reference data of Ghia *et al.* [1].

2.1 Questions

2.1.1 How would you describe the flow field obtained what kinds of features can be distinguished?

The resulting flow field is characteristic of the lid-driven cavity at low Reynolds number. The motion of the top wall generates a large primary recirculation cell that occupies most of the cavity. The core of this primary vortex is located slightly below the moving lid and shifted towards the downstream corner, where the fluid has been accelerated by the lid before turning downwards.

Near the solid boundaries, thin shear layers form and the streamlines are almost parallel to the walls, reflecting the no-slip condition. In addition to the main vortex, small secondary vortices appear in the bottom corners. These corner eddies are a typical feature of the cavity flow and become more clearly resolved as the mesh is refined.

Overall, the flow field shows a dominant primary vortex, secondary corner vortices, and well-developed boundary layers along the walls, which are all consistent with the canonical lid-driven cavity flow at $\mathrm{Re} = 100$.

2.1.2 How does the flow develop in time?

At $t = 0$ the fluid is at rest and the lid is impulsively set into motion. Initially, the effect of the moving wall is confined to a thin shear layer near the top boundary, where a strong horizontal velocity gradient develops. This shear layer is then convected towards the right corner and down along the side wall, gradually inducing a recirculating motion in the interior of the cavity.

As time progresses, vorticity diffuses from the walls into the bulk flow, the primary vortex grows in size and strength, and its center moves towards its final position slightly below and downstream of the lid. The horizontal and vertical velocity profiles along the cavity center lines evolve smoothly from the initial near-zero state towards the characteristic steady shapes, with the largest changes occurring during the first few pass-by times.

Secondary vortices in the bottom corners develop more slowly: they are barely visible at early times but become clearly identifiable as the solution approaches steady state. By $t \approx 10$ s the

centerline velocity profiles at different times nearly collapse, and the flow field can be considered quasi-steady.

2.1.3 How does increasing the mesh resolution affect the results? Comment on the grid-independent nature of the results?

Increasing the mesh resolution has a clear impact on the quality of the solution. On the coarsest grid (10×10), the centerline velocity profiles deviate noticeably from the benchmark data, especially near the velocity extrema and close to the walls. The primary vortex is relatively diffuse and the small corner vortices are only weakly represented.

When the mesh is refined to 40×40 and further to 80×80, the boundary layers and corner vortices become much better resolved and the numerical diffusion is reduced. This is reflected in the centerline profiles, where the 40×40 and 80×80 results almost collapse onto each other and agree closely with the reference data of Ghia *et al.* Over most of the domain the difference between these two finest grids is very small compared to the discrepancy of the coarsest grid.

Since further refinement from 40×40 to 80×80 produces only minor changes in the velocity profiles and in the overall vortex structure, the solution can be considered approximately *grid-independent* at this resolution for $\mathrm{Re} = 100$.

References

[1] U. Ghia, K. Ghia, C. Shin, High-re solutions for incompressible flow using the navier-stokes equations and a multigrid method, Journal of Computational Physics 48 (3) (1982) 387–411. doi:https://doi.org/10.1016/0021-9991(82)90058-4.
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